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Date: 05/03/2026

Title: Foundations of Security Operations and Threat Monitoring

Introduction:

Building strong capabilities in cybersecurity operations begins with a solid understanding of how modern security monitoring environment function. A **Security Operations Center** (SOC) is the core of an organization's defensive strategy, where analysts continuously monitor systems, analyze logs, detect suspicious activity, and respond to incidents in real time.

Developing expertise in this field requires both theoretical knowledge and hands-on practice with real security tools, frameworks, and datasets. By learning SOC operations, log management, monitoring technologies, and incident response workflows, I gained the ability to detect threats, investigate anomalies, and protect systems against evolving cyber-attacks.

This foundation also introduces essential concepts such as the CIA triad, threat intelligence integration, and Defense-in-Depth strategies, preparing analysts to operate effectively in real-world security environments.

Part I. General Introduction and Internship Context

Section 1. Introduction

1. General Background

1.1. Cybersecurity in modern organizations

In modern organizations, cybersecurity has become an essential component of daily operations because information systems are constantly exposed to a wide range of threats. Companies, institutions, and service providers rely heavily on digital infrastructure to store data, manage communication, support business processes, and deliver services.

As a result, any compromise affecting these systems can lead to data loss, service disruption, financial damage, reputational harm, and legal consequences. Cyber threats such as unauthorized access, malware infections, phishing attacks, brute-force attempts, privilege abuse, and network intrusions have therefore made security monitoring and defensive operations a major priority.

The growth of interconnected devices, cloud services, remote access, and online platforms has also increased the attack surface of organizations. This means that security can no longer be limited to preventive controls only; it must also include the capacity to detect suspicious activities, investigate abnormal behaviour, and respond rapidly to incidents. In this context, cybersecurity is not only a technical necessity but also an operational and strategic requirement for organizational resilience.

1.2. Importance of Security Operations Centers (SOC)

A **Security Operations Center** (SOC) plays a central role in the protection of modern information systems. A SOC is a dedicated function or team responsible for continuously monitoring security events, detecting potential threats, analyzing alerts, investigating suspicious activities, and supporting incident response. Its purpose is to provide visibility over the security posture of an organization and to reduce the time between the occurrence of a malicious event and its detection and containment.

The importance of a SOC lies in its proactive and reactive capabilities. On one hand, it enables continuous surveillance through logs, alerts, dashboards, and detection tools. On the other hand, it provides a structured process for triage, investigation, response, and documentation. In practical terms, a SOC helps transform raw technical data into actionable security insight. Without such a function, security events may remain unnoticed, response may be delayed, and the impact of incidents may become more severe. For this reason, understanding SOC operations is fundamental for any cybersecurity intern or analyst involved in monitoring and incident handling activities.

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2. Internship Context

2.1. Host organization and department

This report was prepared as part of my internship in the field of cybersecurity within **CyArt Tech LLP, 2-38K, Patel Vas, At Zak Village, Dehgam, Gandhinagar, Gujarat, 382305 India** specifically in the **SOC Department**. The internship environment provided exposure to core concepts and practices related to security operations, monitoring, detection, and incident handling.

The assigned work was designed to help build foundational knowledge of SOC activities while also introducing practical tasks linked to real-world defensive security operations.

The department's work context emphasized the importance of visibility, log analysis, event correlation, alert management, and operational documentation. This internship task was therefore aligned with the objectives of understanding how modern SOC teams detect suspicious events, use monitoring tools, and document their findings in a structured and professional manner.

2.2. Role during the internship

During this internship, I studied and practice the theoretical and technical foundations of **Security Operations Center** activities, with a particular focus on security monitoring, log management, detection logic, and incident response basics. My work involved learning how security analysts interpret events, how logs are collected and reviewed, how dashboards assist in monitoring, and how alert rules can be configured to identify suspicious behaviour.

In addition to the theoretical learning component, I was expected to perform guided laboratory exercises in a small SOC-style environment. These activities allowed me to move beyond conceptual understanding and begin applying the principles of monitoring and detection using practical tools and test systems. My role therefore combined learning, experimentation, observation, and technical documentation.

2.3. Purpose of the assigned task

The purpose of the assigned task was to develop a clear understanding of SOC fundamentals and to explore, through both theory and practice, the operational workflow of security monitoring and incident handling. More specifically, the task aimed to introduce the essential building blocks of SOC work, including the analysis of logs, the use of SIEM-related tools, the creation of dashboards, the configuration of alerts, and the documentation of security events.

This assignment was also intended to help bridge the gap between academic knowledge and operational cybersecurity practice. By studying the concepts and then applying them in a controlled laboratory environment, the task provided practical insight into how security events are detected, reviewed, escalated, and reported. As a result, it served as an introductory but important step in understanding the daily responsibilities of SOC analysts and the broader logic of defensive security operations.

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3. Problem Statement

3.1. Need for continuous monitoring

Modern information systems generate a very large volume of events from endpoints, servers, network devices, applications, and user activities. Within this mass of data, only a limited number of events may actually indicate malicious behaviour or policy violations, yet these events can be critical. Because threats can occur at any time and may remain hidden among normal system activity, continuous monitoring is necessary to maintain visibility over the security posture of an environment.

Continuous monitoring allows analysts to identify unusual login attempts, suspicious traffic patterns, abnormal service creation, malicious processes, and other indicators

of compromise in near real time. Without such monitoring, organizations risk discovering attacks too late, after significant damage has already occurred.

Therefore, a structured and continuous observation capability is essential for early detection, informed investigation, and timely response.

3.2. Challenges in detecting and responding to threats

Although continuous monitoring is necessary, effective threat detection and response remain challenging for several reasons. First, security tools generate a large number of alerts, many of which may be false positives or low-priority events. Analysts must therefore distinguish meaningful security incidents from routine activity and background noise. Second, logs often come from multiple sources and may use different formats, which makes collection, normalization, and correlation more complex.

Another challenge lies in the interpretation of events. A single failed login or suspicious process may not be enough to confirm malicious activity, but when correlated with other indicators, it may reveal an attack pattern. This requires both technical visibility and analytical judgment. In addition, response actions must be structured and documented properly in order to ensure traceability, consistency, and lessons learned. These challenges demonstrate why SOC work requires not only tools, but also methodology, discipline, and practical understanding of detection workflows.

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4. Objectives of the Report

4.1. General objective

The general objective of this report is to present the theoretical study and practical exploration carried out during the internship on Security Operations Center fundamentals, security monitoring, log analysis, detection logic, and incident response basics. The report aims to show how these concepts were understood and applied in a laboratory environment in order to build foundational skills relevant to SOC operations.

4.2. Specific objectives

The specific objectives of this report are as follows:

- To explain the role and importance of a Security Operations Center in modern cybersecurity operations.
- To study the main concepts related to security monitoring, log management, and incident response.
- To understand the use and purpose of key security tools such as SIEM platforms, IDS/IPS solutions, vulnerability scanners, and endpoint visibility tools.
- To perform practical exercises related to event analysis, monitoring dashboards, and detection rule logic.
- To document the methodology, observations, and learning outcomes resulting from the assigned task.
- To connect theoretical SOC concepts with hands-on laboratory activities in a controlled environment.

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5. Methodology of the Work

5.1. Theoretical study approach

The first part of the work was based on a theoretical study of SOC-related concepts and practices. This included reviewing the purpose of a SOC, the stages of security operations, the fundamentals of log management, the principles of incident response, and the role of monitoring and detection tools in defensive cybersecurity.

The study also involved understanding common security terminology and concepts such as the CIA triad, threats, vulnerabilities, risks, Defense-in-Depth, and zero trust. This theoretical approach was necessary to establish a conceptual foundation before moving to practical work. It helped clarify the logic behind security operations and provided the knowledge required to interpret the technical exercises performed later in the lab environment.

5.2. Practical and lab approach

The second part of the work consisted of practical activities carried out in a laboratory setup designed to simulate a small monitoring and testing environment. The lab architecture used two laptops.

The first laptop hosted a VirtualBox environment containing **Kali Linux**, **Ubuntu**, and **Metasploitable** virtual machines. This setup supported security testing, attack simulation, and the generation or observation of system and network events.

The second laptop was used for accessing the **Kibana dashboard** and conducting technical research, which helped separate the monitoring and analysis interface from the main testing environment.

This practical setup allowed the study of security events from both an offensive-testing and defensive-monitoring perspective. Kali Linux provided a platform for security tools and controlled testing, Metasploitable served as a deliberately vulnerable target machine for safe experimentation, and Ubuntu supported related services and configurations. Kibana was used as a visualization and analysis interface to review collected data, monitor events, and explore dashboard-based security insights.

5.3. Documentation and reporting approach

The work was documented using a structured reporting approach combining theoretical explanation, practical description, and analytical reflection. Each major concept studied was connected, where possible, to a corresponding practical activity or observation. The documentation process aimed to record not only what was studied or performed, but also why it was relevant in SOC operations and what learning outcomes emerged from the task.

This reporting approach was chosen to produce a clear and professional internship document. It allows also to follow the progression from conceptual understanding to hands-on application, while highlighting the relevance of monitoring, detection, and documentation in cybersecurity operations. In this way, the report serves both as evidence of completed work and as a learning record of the internship experience.

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6. Structure of the Report

This report is organized into several parts in order to present the work in a logical and progressive manner.

The first part introduces the general background of cybersecurity, the importance of SOC operations, the internship context, the problem statement, the objectives of the report, and the methodology used.

The second part presents the theoretical foundations related to SOC fundamentals, security monitoring, log management, security tools, and incident response.

The third part describes the practical activities carried out in the laboratory environment, including monitoring-related tasks, event analysis, dashboard usage, and alert logic exploration.
The fourth part presents the results, observations, challenges encountered, and key learning outcomes.

Finally, the report concludes with a general summary of the work, recommendations, references, and appendices containing supporting evidence such as screenshots, configurations, or sample outputs.

Part II. Theoretical Foundations of SOC Operations

Section 2. SOC Fundamentals and Core Security Concepts

1. SOC Fundamentals and Operations

A **Security Operations Center** (SOC) is a centralized function responsible for monitoring, detecting, analyzing, and responding to cybersecurity events within an organization's information systems.

Its main purpose is to maintain continuous visibility over the security posture of the environment and to reduce the time required to identify and react to suspicious

activities. In practice, a SOC does not simply collect alerts from tools; it transforms technical events into actionable information that can support investigation, containment, and recovery.

The SOC plays an essential role in defensive security operations because modern infrastructures generate a large volume of logs and events from endpoints, servers, applications, and network devices. Among these events, some may indicate normal operational behaviour, while others may reveal unauthorized access attempts, malware activity, misconfigurations, insider misuse, or external attacks. The SOC is therefore responsible for continuously reviewing security data, correlating events from multiple sources, and deciding which alerts require further action.

The main functions of a SOC include continuous monitoring, alert triage, event investigation, incident escalation, reporting, and coordination with response teams. Monitoring involves watching dashboards, logs, and detection rules for suspicious behaviour.
Triage consists of assessing alerts according to severity, confidence, and potential impact. Investigation requires analysts to examine evidence from different data sources in order to determine whether an event represents a real threat. When an incident is confirmed, the SOC supports or initiates response actions and ensures that the incident is documented properly.

Several roles are commonly found within a SOC.

Tier 1 analysts are generally the first line of defense. They monitor alerts, review dashboards, perform initial triage, and escalate suspicious cases when necessary.

Tier 2 analysts handle deeper investigations, correlate multiple indicators, validate incidents, and recommend containment actions.

Tier 3 analysts usually deal with advanced analysis, complex threats, malware behaviour, threat hunting, and detection improvement.

The SOC manager supervises daily operations, team coordination, workflow performance, and reporting. In some organizations, **threat hunters** proactively search for hidden or undetected malicious activity by analyzing patterns and behaviours that may not yet have triggered alerts.

Figure 1: SOC Team hierarchy

In a practical SOC environment, these roles work together to ensure that threats are identified and handled in a structured way. This coordination is important because cybersecurity events often evolve quickly, and delayed analysis may increase the impact of an attack. For this reason, the SOC is not only a technical capability but also an operational framework that combines people, processes, and tools in order to support organizational security.

2. Basic Security Concepts

A strong understanding of core security concepts is necessary for effective SOC work because analysts must interpret events according to their impact on systems, users, and business operations.

One of the most fundamental models in cybersecurity is the **CIA triad**, which refers to **confidentiality, integrity,** and **availability**. Confidentiality means that information must only be accessible to authorized individuals or systems. Integrity means that information must remain accurate, complete, and protected from unauthorized alteration. Availability means that systems, services, and data must remain accessible when needed by legitimate users. These three principles form the basis for understanding the objectives of cybersecurity controls and the consequences of security incidents.

Figure 2: CIA Triad Figure

Another important distinction in cybersecurity is the difference between a **threat**, a **vulnerability**, and a **risk**.
A threat is any actor, event, or condition that has the potential to cause harm to a system or data. This may include attackers, malware, insider misuse, or even accidental errors.

A vulnerability is a weakness or flaw in a system, configuration, application, or process that could be exploited by a threat.
Risk refers to the likelihood that a threat will exploit a vulnerability and the resulting impact on the organization.

In other words, a vulnerability alone does not necessarily create damage, but when combined with a relevant threat, it produces risk.

SOC analysts also need to understand the principle of **Defense-in-Depth**. This approach involves applying multiple layers of security controls rather than depending on a single protective mechanism. For example, an organization may combine firewalls, endpoint protection, network monitoring, access controls, logging, alerting, and user awareness measures.

The idea is that if one layer fails, other layers can still prevent, detect, or reduce the impact of an attack. Defense-in-Depth is highly relevant in SOC work because monitoring and response are often the layers that identify threats which were not stopped by preventive controls.

A related modern concept is **zero trust**. Zero trust is based on the idea that no user, device, or system should be trusted automatically, even if it is inside the internal network. Access should be granted only after verification of identity, device condition, context, and policy compliance. In practical terms, zero trust promotes strict authentication, least-privilege access, segmentation, and continuous validation. For SOC operations, this concept is important because it supports more granular monitoring, clearer anomaly detection, and better visibility into access patterns that may indicate compromise or misuse.

Together, these concepts help analysts understand why monitoring matters and how security events affect the protection of information assets. They also provide the conceptual language needed to assess incidents, communicate findings, and support security decision-making.

3. Security Monitoring Basics

Security monitoring is the continuous observation and analysis of events generated by systems, networks, applications, and users in order to identify suspicious behaviour, policy violations, and signs of compromise.

Its primary objective is to provide visibility into security-relevant activities so that analysts can detect threats early and respond before significant harm occurs. In a SOC environment, monitoring is one of the most important daily functions because it serves as the main mechanism through which abnormal activities are identified. The monitoring process involves collecting data from multiple sources and analyzing it for patterns that may indicate malicious or unauthorized behaviour. These patterns may include **repeated failed login attempts, unusual network traffic, unexpected service creation, privilege escalation, malware execution, unauthorized outbound communication,** or **access outside normal working hours**. By reviewing such indicators, analysts can identify events that deserve further investigation.

Security monitoring is not limited to obvious attacks. It also aims to detect misconfigurations, weak controls, insider misuse, and policy violations that may create security exposure.

For example, a large number of authentication failures may suggest a brute-force attempt, while the execution of a new service on a host may indicate persistence activity or unauthorized software installation. Therefore, monitoring helps transform raw technical data into operational awareness.

An important challenge in monitoring is managing the quality of alerts. **A false positive** occurs when an alert indicates suspicious activity that later turns out to be benign. **A false negative** occurs when malicious activity is not detected at all.

Both problems are significant. Too many false positives waste analyst time and create alert fatigue, while false negatives allow real attacks to remain hidden. For this reason, detection rules and monitoring logic must be tuned carefully.

Another key performance indicator is **Mean Time to Detect** (MTTD), which refers to the average time required to identify a threat after it has occurred. A lower MTTD generally indicates a stronger monitoring capability because suspicious activity is recognized more quickly.

In SOC operations, reducing detection time is critical because early identification often limits the impact of incidents and improves the effectiveness of response actions.

Overall, security monitoring provides the operational bridge between raw system activity and incident response. It enables analysts to discover what is happening in the environment, determine what requires attention, and prioritize actions based on evidence.

Section 3. Log Management and Security Tools

1. Log Management Fundamentals

Logs are records generated by systems, devices, applications, and security tools that document events occurring within an environment. They may contain information about user authentication, process execution, file access, network communication, system errors, service changes, application requests, and many

other activities. In SOC operations, logs are essential because they provide the evidence needed to detect suspicious activity, reconstruct incidents, validate alerts, and understand how events unfolded over time.

The value of logs lies in their ability to **reveal both normal and abnormal behaviour.**

When reviewed correctly, logs can help identify brute-force attempts, malware execution, unauthorized access, configuration changes, suspicious connections, or unusual service activity. In addition, logs are important for compliance, auditing, forensic analysis, and post-incident review. Without logs, analysts would have very limited visibility into what happened on a system or network.
Common **log sources** include **Windows Event Logs, Syslog, web server logs, firewall logs, antivirus** or **EDR telemetry, authentication records, DNS logs,** and **application logs.**

Windows Event Logs are useful for tracking user logins, account activity, service creation, and system changes on Windows systems.
Syslog is commonly used in Linux systems, network devices, and servers to record operational and security-related events.
Web logs capture requests made to web servers and can reveal suspicious traffic patterns, scanning behaviour, or malicious requests.

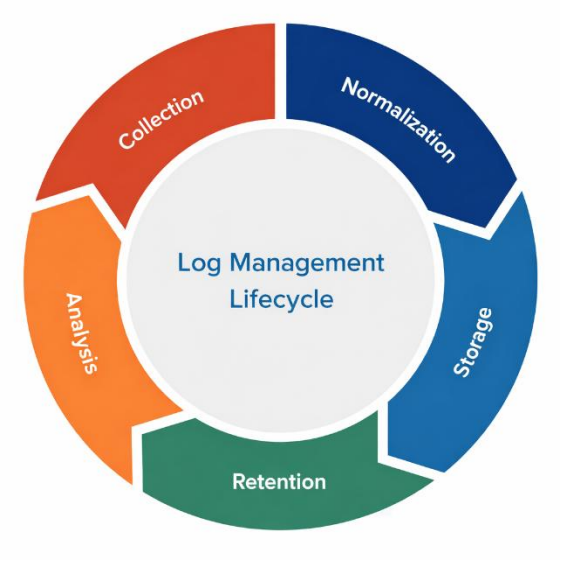

Figure 3: Log Management Lifecycle

Log management follows a lifecycle that generally includes **collection, normalization, storage, retention,** and **analysis.**

Collection refers to gathering logs from different sources using agents, forwarders, or log ingestion pipelines.
Normalization is the process of converting logs from different formats into a consistent structure so that they can be searched and correlated more easily.
Storage involves saving the logs in a centralized location such as a SIEM or log management platform.
Retention refers to keeping logs for an appropriate period according to operational, investigative, or compliance needs.
Analysis is the stage where logs are searched, filtered, correlated, and interpreted to identify meaningful patterns or incidents.

A strong log management process improves SOC effectiveness because it ensures that relevant data is available, understandable, and searchable when analysts need

it. Good log management supports both real-time monitoring and retrospective investigation, making it a foundational component of modern security operations.

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2. Security Tools Overview

SOC operations rely on several categories of security tools, each designed to provide visibility over a specific part of the environment. These tools do not replace analysts, but they increase the speed, depth, and consistency of monitoring and investigation activities.

A **Security Information and Event Management** (SIEM) platform is used to collect, centralize, search, correlate, and visualize logs from multiple sources. SIEM tools help analysts review security data in one place, create dashboards, define alerting rules, and investigate suspicious events. They play a central role in SOC work because they bring together information from endpoints, servers, network devices, and applications.

Endpoint Detection and Response (EDR) tools focus on activity occurring on endpoints such as laptops, desktops, and servers. These tools monitor processes, files, registry changes, command execution, user behaviour, and other host-level actions. Their purpose is to identify suspicious behaviour directly on the endpoint, support rapid investigation, and in some cases provide containment actions such as isolating a machine or terminating a malicious process.

Intrusion Detection Systems (IDS) and **Intrusion Prevention Systems** (IPS) are used to monitor network traffic for malicious signatures, suspicious patterns, or policy violations. An IDS detects suspicious activity and generates alerts, while an IPS can also take preventive action, such as blocking traffic.

These tools are important for identifying network-based threats, scans, exploit attempts, or abnormal communications that may not be visible from host logs alone.

Vulnerability scanners are used to identify weaknesses in systems, applications, or configurations before they are exploited by attackers. They help organizations understand their exposure by detecting missing patches, insecure settings, known vulnerabilities, and outdated software.
Although vulnerability scanners are not primarily monitoring tools, they are highly relevant to SOC operations because they help explain where risk exists and which assets may be easier targets.

Endpoint visibility tools, such as system query frameworks, provide analysts with detailed information about the internal state of hosts. They may reveal running processes, open ports, installed software, active users, scheduled tasks, or configuration details. This visibility is valuable during investigations because it allows analysts to verify whether suspicious activity is currently present on a machine and whether persistence or misuse may exist.

Together, these tool categories create a broader monitoring ecosystem. SIEM provides centralized visibility, EDR adds host-level detection, IDS/IPS watches network traffic, vulnerability scanners expose weaknesses, and endpoint visibility tools support investigation and validation. In a SOC, the strength of monitoring often depends not on one tool alone, but on how these tools complement one another.

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3. Documentation Standards in Security Operations

Documentation is a critical component of SOC operations because security work must be accurate, repeatable, and understandable by different team members. Even when detection and investigation are technically successful, the value of that work is reduced if findings are not documented clearly. Good documentation ensures that incidents can be tracked, reviewed, escalated, and learned from in a structured manner.

One important type of documentation is the **runbook**. A runbook is a practical guide that explains how to handle a specific category of alert or incident. It provides step-by-step instructions so that analysts can respond consistently and efficiently. **Standard Operating Procedures** (SOPs) are broader documents that define routine operational processes, responsibilities, and expected workflows within the SOC. Both runbooks and SOPs are important because they reduce confusion, support standardization, and improve response quality.

Incident reports are used to document the details of a security event or confirmed incident. Such reports usually include the date and time of the event, systems or users involved, source and destination information, indicators observed, actions taken, impact assessment, and recommendations. Incident reporting is important for communication, accountability, management review, and future improvement of detection and response procedures.

Post-incident reviews, sometimes called lessons learned or post-mortems, are also essential. These reviews analyze what happened, how the incident was detected, how well the response worked, and what improvements are needed. They help organizations strengthen controls, refine alerts, update runbooks, and reduce the likelihood of similar issues in the future.

In SOC operations, technical findings must therefore be recorded in a disciplined and structured way. Documentation transforms isolated technical actions into organizational knowledge. It supports continuity between analysts, strengthens investigation quality, and contributes to the overall maturity of the security function.

Section 4. Incident Response and Security Operations Workflow

1. Security Operations Workflow

The security operations workflow describes how analysts move from the initial detection of suspicious activity to the final response and documentation of an event. This workflow is important because it provides a structured path for handling alerts efficiently and consistently. In SOC operations, the general chain can be summarized as detection, triage, investigation, and response.

Detection is the stage where a potentially suspicious event is identified through logs, dashboards, alert rules, endpoint tools, or network monitoring systems. At this point, the event is not always confirmed as malicious; it is simply recognized as something that deserves attention. Detection may be based on a known signature, a threshold rule, a behavioural anomaly, or analyst observation.

Triage is the process of evaluating the detected event in order to determine its urgency, credibility, and possible impact. During triage, the analyst checks whether the alert is likely to be a false positive, whether it affects a critical asset, whether similar alerts have appeared before, and whether escalation is necessary. Triage is important because SOC environments often generate many alerts, and analysts must prioritize their effort effectively.

Investigation begins when an event appears suspicious enough to require deeper analysis. During this phase, the analyst correlates logs, reviews host and network evidence, checks user and asset context, and looks for indicators of compromise or attack progression. The goal is to understand what happened, how it happened, which systems are affected, and whether the event represents a real security incident.

Response is the stage where action is taken based on the findings of the investigation. Depending on the situation, response may involve isolating an endpoint, blocking an IP address, disabling an account, removing malicious files, escalating to another team, or launching formal incident response procedures. Response must also be documented carefully so that the organization can track the event and improve its defenses in the future.

This workflow shows that SOC work is not limited to receiving alerts. It is a disciplined process of filtering, validating, analyzing, and acting on security information. The effectiveness of the SOC depends not only on the quality of tools, but also on the consistency of this workflow.

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2. Incident Response Basics

Incident response is the structured process used to manage and contain cybersecurity incidents in order to reduce damage, restore operations, and improve future readiness. It provides a formal approach for handling events that have moved beyond simple monitoring and are confirmed, or strongly suspected, to be security incidents.

The first stage is **preparation**. In this stage, the organization establishes policies, defines roles, prepares tools, creates communication procedures, trains personnel, and develops runbooks or playbooks. Preparation is important because effective response depends heavily on decisions and resources that must already be in place before an incident occurs.

The second stage is **identification**. This is the point at which suspicious activity is reviewed and confirmed as an incident or potential incident. Analysts determine what has happened, what systems may be involved, and whether the observed evidence justifies escalation. Identification is closely linked to monitoring and investigation because it relies on logs, alerts, and contextual analysis.

The third stage is **containment**. Once an incident is confirmed, actions are taken to limit its spread or reduce its immediate impact. Examples include isolating a compromised machine, disabling a user account, blocking malicious communication, or restricting access to affected systems. Containment must balance speed with caution, because overly aggressive actions may disrupt business operations or destroy useful evidence.
The fourth stage is **eradication**. This involves removing the root cause of the incident, such as malware, unauthorized accounts, malicious scripts, or exploited vulnerabilities. The purpose of eradication is to eliminate the threat from the environment so that it cannot continue or reappear immediately.

The fifth stage is **recovery**. During recovery, systems are restored to normal operation in a controlled manner. This may involve restoring services, validating system integrity, monitoring for signs of recurrence, and ensuring that affected assets are safe to return to production use.

The final stage is **lessons learned**. After the incident is handled, the organization reviews what happened, how it was detected, how effective the response was, and what should be improved. This step is essential because it turns operational experience into stronger procedures, better alerts, improved controls, and greater overall resilience.

For a SOC analyst, understanding this lifecycle is important because monitoring and investigation are only part of the larger incident handling process. Each stage contributes to reducing impact and improving the organization's future ability to respond.

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3. Use of Frameworks and Standards

Frameworks and standards help structure SOC operations and incident response by providing recognized methods, terminology, and models. They are important because they make security work more consistent, repeatable, and aligned with industry practices.

One widely used reference is the **NIST incident response model**. This model provides a clear structure for incident handling through stages such as preparation, detection and analysis, containment, eradication and recovery, and post-incident activity. It helps organizations organize responsibilities, establish procedures, and evaluate their response maturity. In a SOC context, the NIST model is useful because it connects monitoring, investigation, and response within a formal operational process.

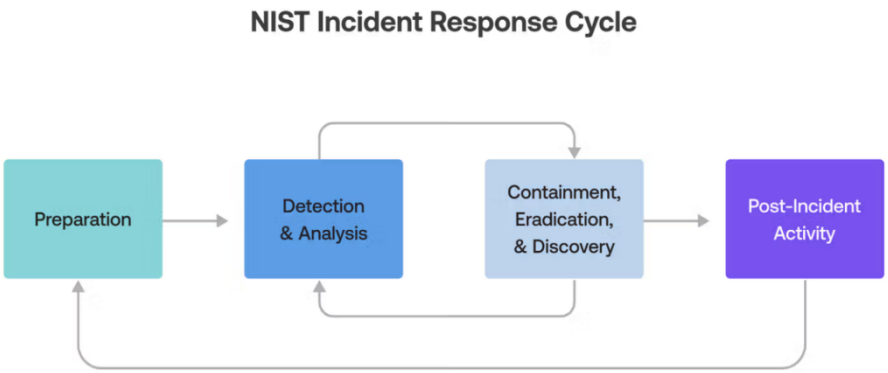

Figure 4: NIST IR lifecycle

Another important framework is **MITRE ATT&CK**. MITRE ATT&CK is a knowledge base that classifies adversary tactics and techniques based on observed attacker behaviour. Instead of focusing only on malware names or isolated alerts, it helps analysts understand what an attacker is trying to achieve, such as initial access, execution, persistence, privilege escalation, defense evasion, credential access, discovery, lateral movement, and exfiltration. This approach is valuable for SOC

teams because it supports better detection engineering, more meaningful investigations, and stronger threat hunting.

In practical monitoring work, MITRE ATT&CK can help analysts map observed events to adversary behaviour. For example, repeated failed logins may relate to credential access attempts, suspicious command execution may indicate execution techniques, and unauthorized service creation may suggest persistence. By using such a framework, analysts can move beyond isolated event review and develop a more strategic understanding of attacker activity.

Overall, frameworks such as NIST and MITRE ATT&CK improve the discipline and maturity of security operations. They provide structure to workflows, strengthen communication, and help analysts interpret events in a way that is operationally useful and aligned with recognized cybersecurity practices.

Part III. Practical Activities Carried Out During the Task

Section 5. Log Analysis and Event Investigation Practice

1. Log Analysis Practice

1.1. Windows Event Viewer analysis

As part of the practical activities, log analysis was studied through the review of security-relevant events in a **Windows** environment using **Event Viewer**. This activity was important because host-level logs provide direct visibility into authentication behaviour, system activity, and administrative changes.

In SOC operations, Windows Event Viewer is a useful starting point for understanding how operating system logs can reveal potential compromise, suspicious user behaviour, or unauthorized changes. The exercise helped illustrate how analysts move from raw event records to the identification of meaningful security indicators during initial monitoring and investigation.

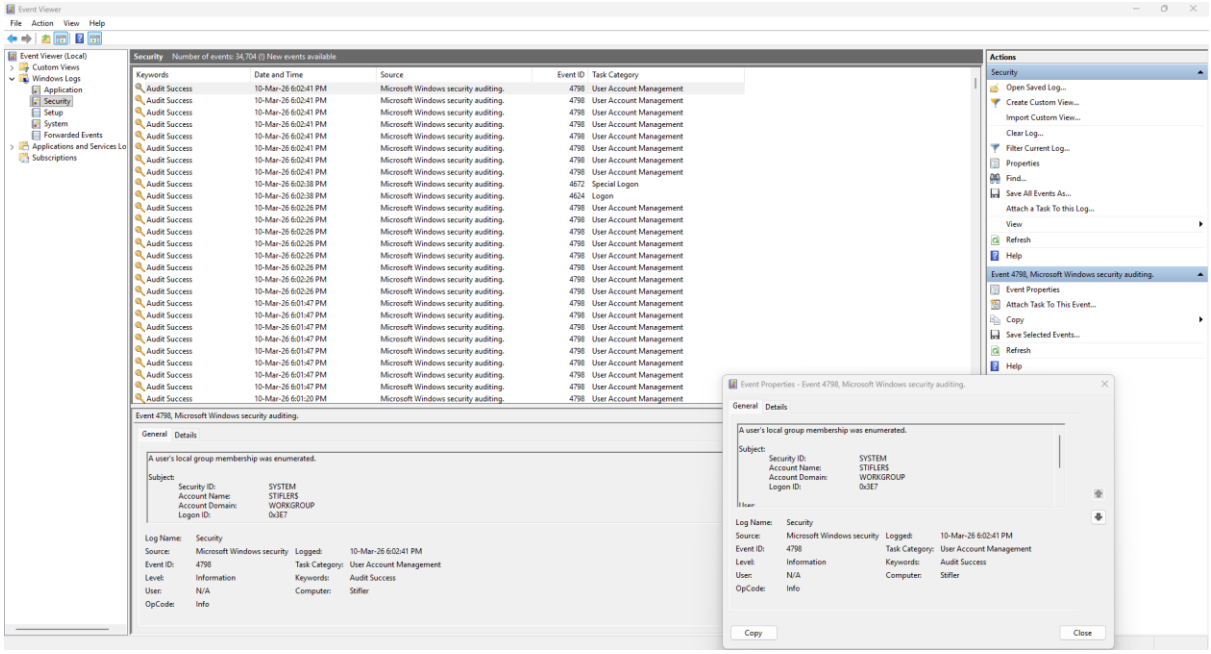

Figure 5: Windows Event Viewer used to review security-related events

Event ID	Meaning
4625	Multiple failed logins
4624	Successful login
4672	Admin privileges assigned
4688	powershell.exe executed
4798	Group enumeration
4720	New user created
1102	Logs cleared

Table 1: Important Event ID for SOC Analyst

1.2. Detection of failed logins (Event ID 4625)

A specific focus of the log analysis practice was the detection of failed login attempts through **Event ID 4625**, which is commonly associated with unsuccessful authentication attempts in Windows systems. This type of event is highly relevant in SOC work because repeated authentication failures may indicate brute-force attempts, unauthorized access trials, password spraying, or user account misuse.

By reviewing such events, an analyst can identify who attempted the login, when it occurred, from which source it originated, and whether the pattern appears isolated or repeated. This activity demonstrated how authentication logs can serve as an early indicator of suspicious behaviour requiring further investigation.

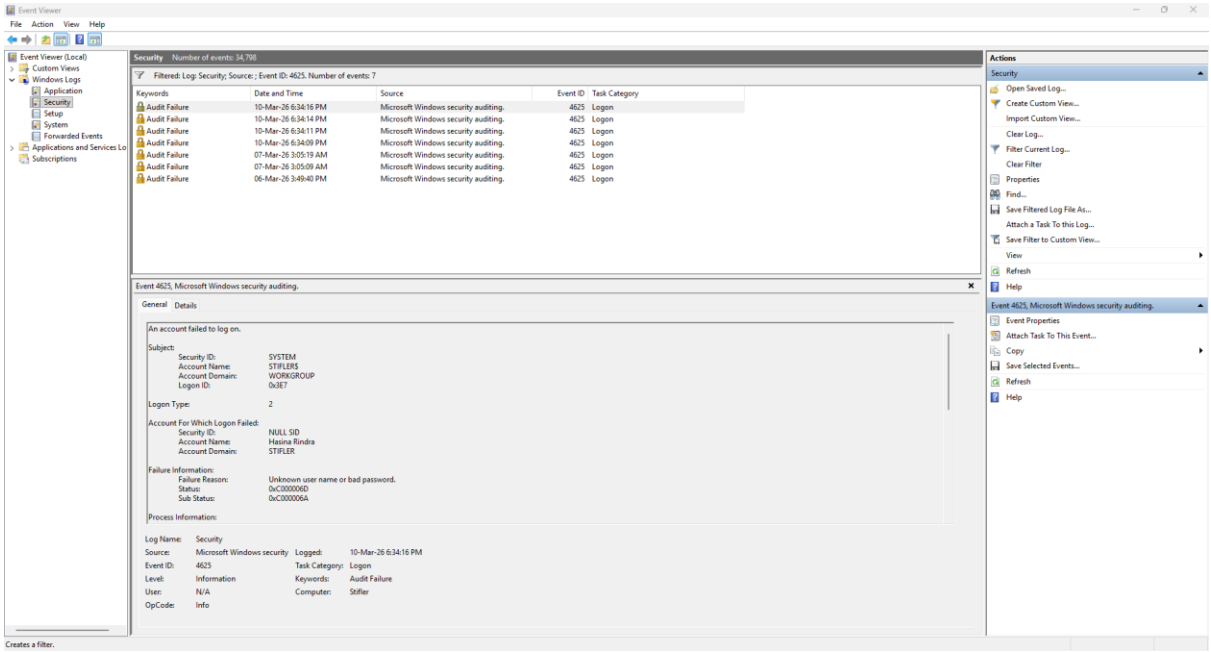

Figure 6: Example of failed login events (Event ID 4625) reviewed

1.3. Detection of new service creation (Event ID 7045)

Another important log analysis task involved reviewing **Event ID 7045**, which records the creation of a new service on a Windows system. This event is significant because attackers may create malicious or unauthorized services in order to establish persistence, execute code automatically, or maintain long-term access to a compromised host.

In a SOC context, the appearance of a newly installed service should be reviewed carefully, especially if the service name, path, or associated account appears unusual. Studying this event helped reinforce the idea that not all suspicious activity appears in network logs alone; host-based event records are also essential for identifying internal changes that may indicate compromise or unauthorized administrative activity.

1.4. Identification of suspicious patterns

Beyond reviewing isolated events, the practical activity also focused on identifying suspicious patterns across multiple records. In SOC work, a single event may not be enough to confirm malicious behaviour, but a repeated or correlated sequence of events may reveal an attack attempt or policy violation.

For example, several failed login attempts occurring in a short period may suggest brute-force behaviour, while the combination of unusual authentication activity and service creation may justify deeper investigation. This exercise helped demonstrate the importance of moving beyond single-event observation toward pattern recognition, correlation, and contextual interpretation, which are essential analytical skills in security monitoring.

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2. Browser History Analysis

2.1. Purpose of browser artifact analysis

Browser history analysis was included as a practical activity to highlight the value of endpoint artifacts in security investigations. In many cases, browser records can provide useful evidence related to phishing, malicious downloads, suspicious URLs, command-and-control access, or user interaction with untrusted websites.

For a SOC analyst or investigator, browser artifacts can help reconstruct user activity, confirm whether a suspicious link was accessed, and support the timeline of a security event. This part of the task showed that incident investigation is not limited to alerts and logs only, but may also involve endpoint-level artifacts that provide additional context.

2.2. Use of forensic tools for URL review

To support browser history analysis, forensic tools may be used to extract, parse, and review stored URL records in a structured manner. Such tools help analysts identify browsing activity more efficiently than manual checking, especially when reviewing large volumes of history entries.

In the context of the assigned task, the use of URL review tools illustrated how forensic support can complement monitoring activities by providing evidence of visited websites, timestamps, and possible links to suspicious content. This reinforces the broader SOC principle that different data sources must often be combined during investigation in order to reach a more reliable conclusion.

2.3. Identification of suspicious or test URLs

The practical value of browser history analysis lies in the ability to identify suspicious or test URLs that may be related to phishing, malicious redirection, or user access to untrusted resources. In a controlled laboratory or learning environment, such analysis may also involve test URLs used to simulate suspicious browsing behaviour.

Reviewing these entries helps demonstrate how analysts can validate whether a user visited a suspicious domain, whether the access matches an alert timeline, and whether the browsing activity supports a larger pattern of compromise. This exercise therefore contributes to the analyst's ability to connect user activity with event-based monitoring evidence.

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3. Event Documentation Practice

3.1. Event documentation template

As part of the practical learning process, event documentation was treated as a key SOC activity rather than a final administrative step. A structured event documentation template helps ensure that relevant information is recorded consistently and can later be reviewed, escalated, or compared with other cases.

In a SOC environment, even a technically valid finding loses value if it is not documented clearly. For this reason, a standard reporting format was used or designed to capture essential event details in a concise and traceable manner.

3.2. Recording date, source IP, event ID, description, and action taken

The documentation practice involved recording key fields such as the event date and time, source IP or originating host, event identifier, short description of the observed activity, and the action taken by the analyst. These fields are important because they transform raw technical observations into structured evidence.

For example, when documenting repeated failed logins, the analyst can record the timestamp, targeted account, event code, likely interpretation, and whether the case was escalated or retained for monitoring. This exercise helped illustrate how documentation supports not only memory and reporting, but also reproducibility and communication between analysts.

3.3. Importance of structured evidence recording

Structured evidence recording is essential because SOC analysis often involves multiple data sources, several analysts, and the possibility of later review by supervisors or incident response teams. Properly recorded evidence supports decision-making, reduces ambiguity, and helps maintain an accurate timeline of what was observed and what actions were taken.

It also contributes to organizational learning, because documented cases can later be used to improve detection rules, refine runbooks, and support post-incident analysis. This part of the exercise emphasized that documentation is not separate from technical work; it is one of the foundations of professional security operations.

Section 6. Monitoring Configuration and Detection Logic

1. Monitoring Dashboard Setup

1.1. Purpose of dashboards in SOC operations

Dashboards are an essential part of SOC operations because they provide analysts with a centralized visual view of security-relevant information. Instead of reading raw logs one by one, analysts can use dashboards to observe trends, prioritize alerts, track event frequency, and identify unusual behaviour more quickly.

In practical terms, dashboards support situational awareness by making patterns easier to detect, such as spikes in failed authentication attempts, repeated alerts from the same source, or concentration of events over a specific time period. This activity helped demonstrate how dashboards transform collected security data into usable operational visibility.

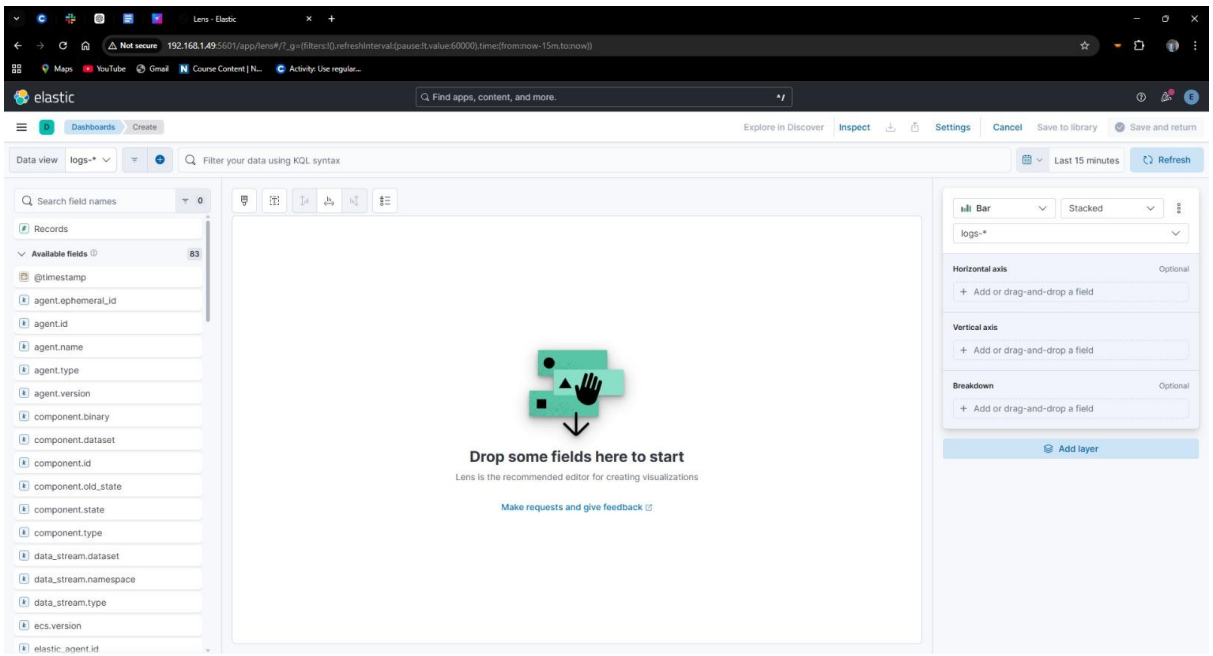

Figure 7: Dashboard interface used to visualize monitored security events

1.2. Dashboard design in Kibana

The dashboard design activity focused on understanding how monitoring interfaces can be configured to display meaningful security information. In tools such as Kibana, dashboards can be customized to show charts, tables, timelines, and alert summaries according to the analyst's needs.

A good dashboard does not simply display data; it organizes information in a way that supports rapid review and decision-making. In the context of this task, dashboard design was approached as an important part of practical SOC workflow, because the usefulness of monitoring depends not only on the data collected but also on how clearly it is presented.

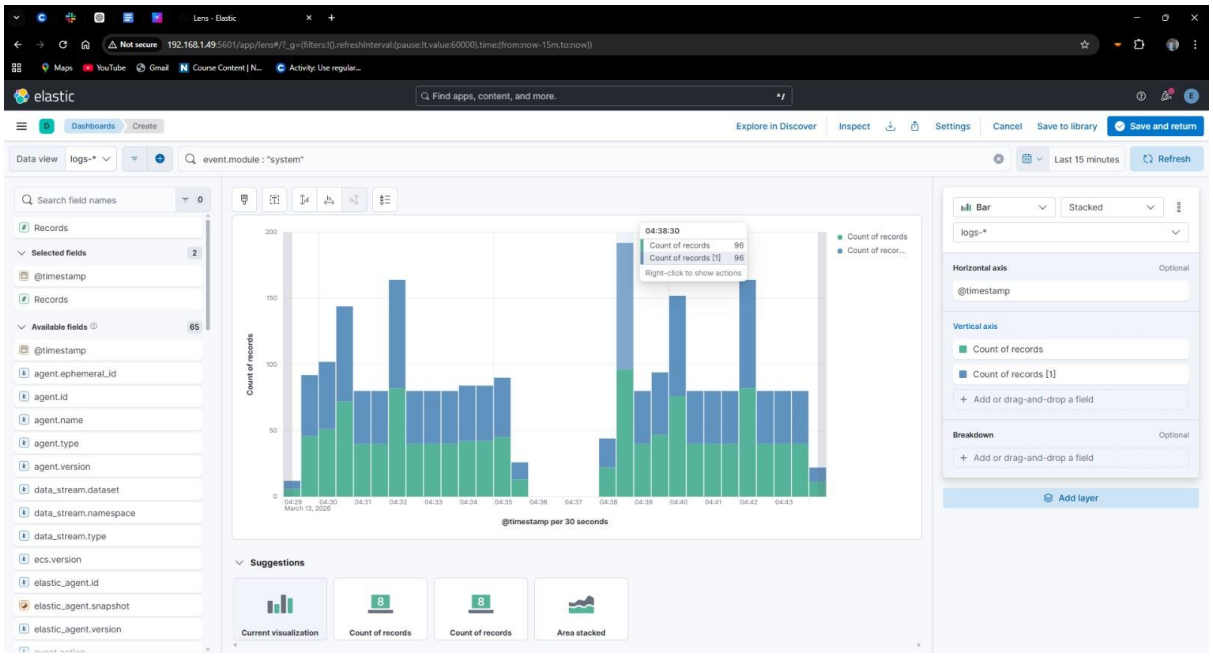

Figure 8: Example of monitoring dashboard layout used for security event visualization

1.3. Visualization of top alert sources

One useful monitoring view consists of identifying the most frequent sources generating alerts or suspicious events. This type of visualization helps analysts quickly recognize which IP addresses, hosts, users, or services are producing the highest volume of activity.

Such information is valuable because a concentration of alerts from the same source may indicate scanning, brute-force attempts, persistent misconfiguration, or an asset requiring closer review. By observing top alert sources through a dashboard, analysts can prioritize investigation and reduce the time needed to identify the most active contributors to suspicious behaviour.

1.4. Visualization of critical events frequency

Another useful dashboard function is the visualization of the frequency of critical or notable events over time. This allows analysts to observe whether a certain event type is rare, recurring, increasing, or concentrated within a particular period.

For example, repeated failed logins, new service creation events, or IDS alerts may become more meaningful when viewed as a trend rather than as isolated records. In SOC operations, frequency-based visualization supports both detection and prioritization by helping analysts distinguish background noise from unusual activity bursts.

---

2. Alert Rule Configuration

2.1. Logic of threshold-based detection rules

Threshold-based detection rules are used in SOC environments to trigger alerts when a specific event occurs more than a defined number of times within a given

time window. This logic is useful because some individual events may be benign when seen once, but suspicious when repeated frequently.

For example, one failed login may be normal, while multiple failed logins in a short period may indicate brute-force behaviour. Studying threshold-based logic helped demonstrate how detection engineering converts raw event repetition into meaningful alerts that can support faster triage and investigation.

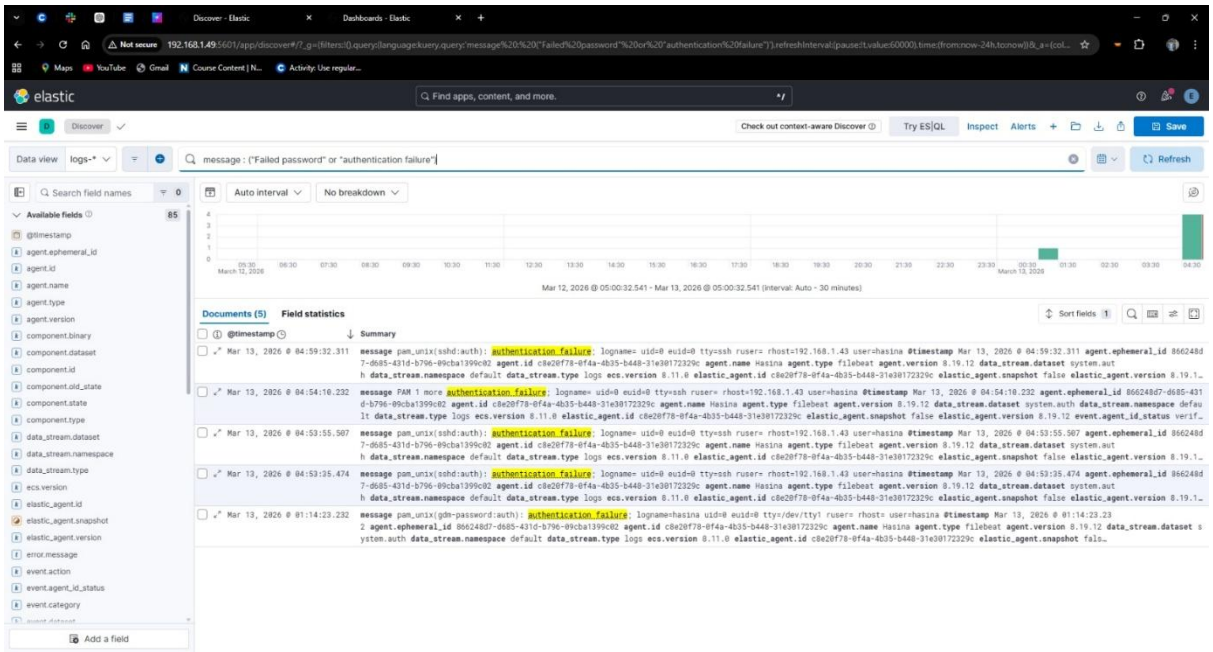

Figure 9: Threshold-based detection logic used for repeated event monitoring

2.2. Rule for repeated failed logins

A specific example of detection logic used in this task was a rule designed to identify repeated failed login attempts within a defined time period. This rule is directly relevant to SOC operations because repeated authentication failures are a common early sign of unauthorized access attempts, brute-force activity, or password misuse.

By setting a threshold, such as several failed logins within a few minutes, the analyst can reduce the need for manual log review and receive an alert only when the

behaviour becomes potentially suspicious. This exercise showed how a common security scenario can be translated into a practical monitoring rule.

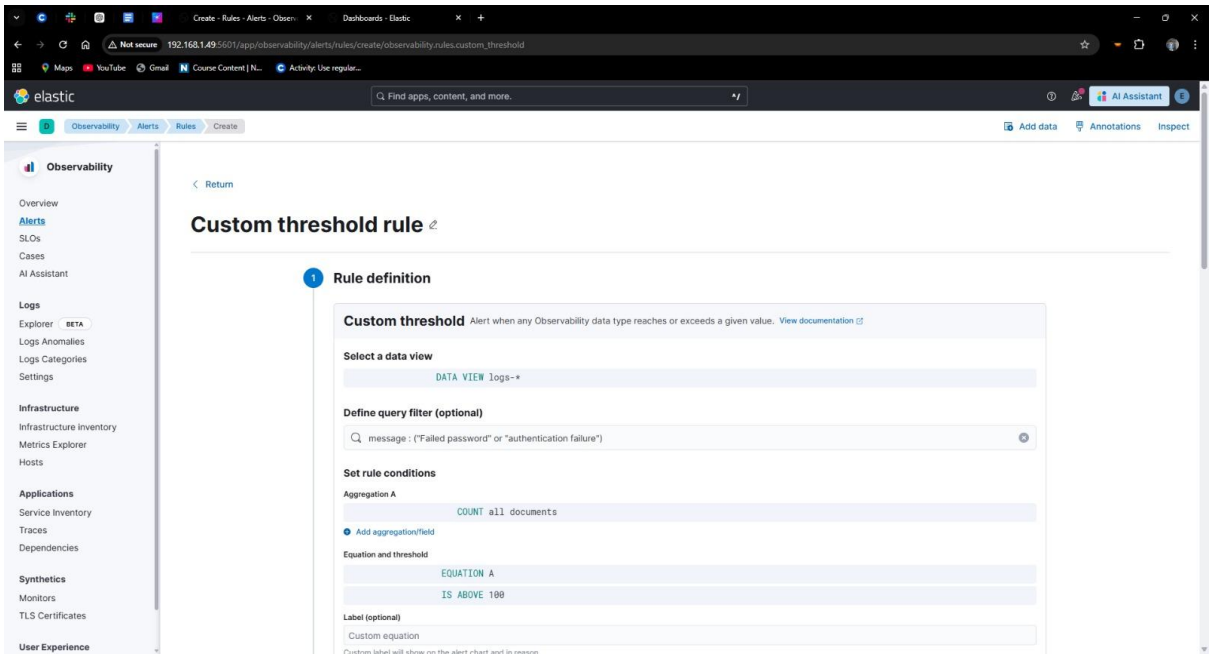

Figure 10: Detection rule configured to identify repeated failed login attempts

2.3. Testing the alert with simulated events

To validate the usefulness of the alert rule, simulated events can be generated in a controlled environment. In this task, the test logic consisted of producing repeated failed authentication attempts or similar trigger conditions in order to verify that the rule behaved as expected. This step is important because a rule should not be considered reliable until it has been tested under realistic conditions. The testing process helps confirm whether the alert is triggered correctly, whether the threshold is suitable, and whether the event appears in the dashboard or alert interface as intended.

2.4. Validation of generated alerts

The final step in rule configuration is alert validation, which consists of confirming that the generated alert is visible, understandable, and operationally useful.

Validation includes checking whether the alert contains the expected event details, whether it appears in the appropriate dashboard or alert list, and whether the information is sufficient for triage and investigation.

This exercise reinforced the idea that an alert is only useful if it provides meaningful context to the analyst. A detection rule that generates unclear or excessive alerts may increase noise rather than improve monitoring effectiveness.

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3. Practical Use of SOC Tools

3.1. Use of SIEM platform

A SIEM platform was used as the central point for collecting, reviewing, and visualizing security-relevant information during the task. Its practical value lies in the ability to centralize logs from different sources, search events more efficiently, build dashboards, and support alerting logic.

In this activity, the SIEM environment helped connect monitoring with investigation by allowing observed events to be reviewed in a more structured and searchable way. This reinforced the understanding that SIEM platforms are not just storage tools, but analytical environments that support SOC workflows from detection to triage.

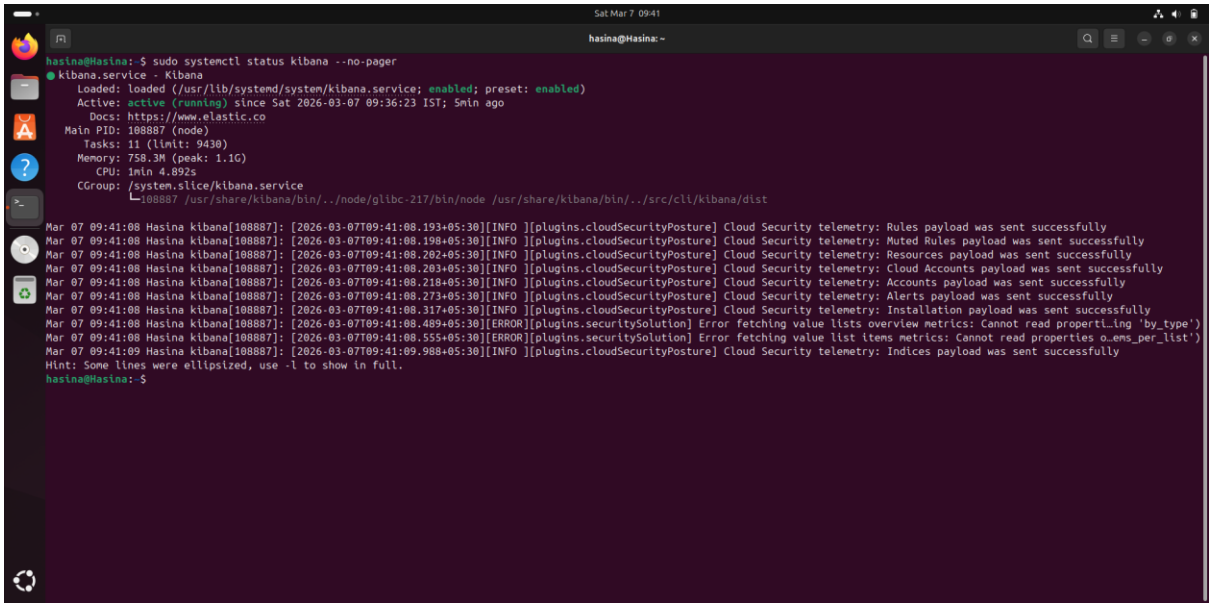

Figure 11: ELK running on Ubuntu on VM

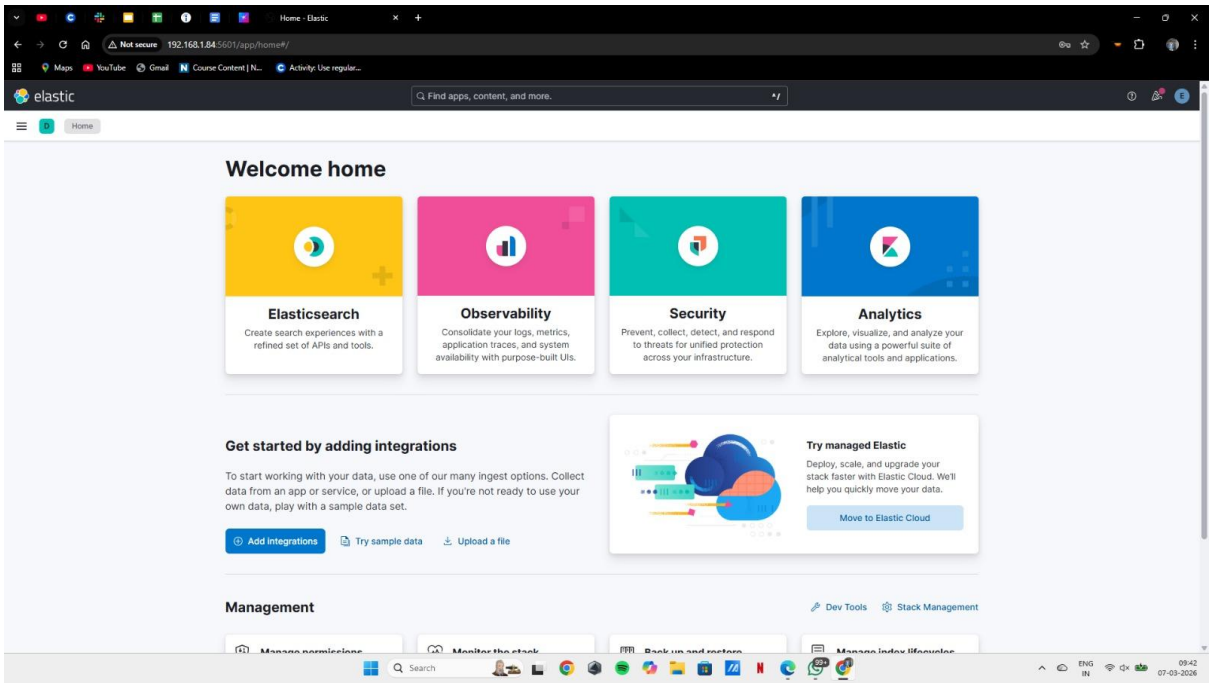

Figure 12: Kibana running on 2nd Windows machine

3.2. Use of Snort / IDS logic

The practical study of IDS logic focused on the role of signature- or rule-based network detection in identifying suspicious traffic patterns. Tools such as Snort are useful in SOC environments because they inspect network traffic and generate alerts when traffic matches known malicious patterns or policy-defined conditions.

Studying IDS logic helped show how network-layer visibility complements host-based logs and SIEM analysis. It also highlighted the importance of rule quality, because detection effectiveness depends on how well signatures and alert logic represent the targeted behaviour.

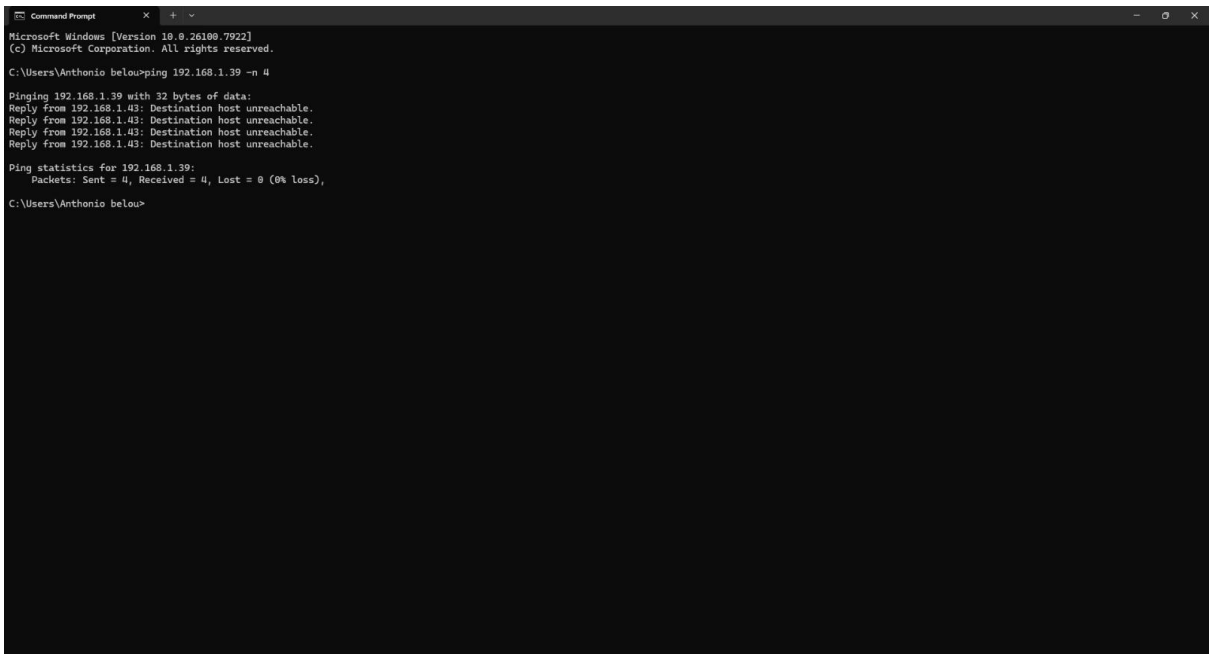

Figure 13: Ping Test Through 2nd Windows machine

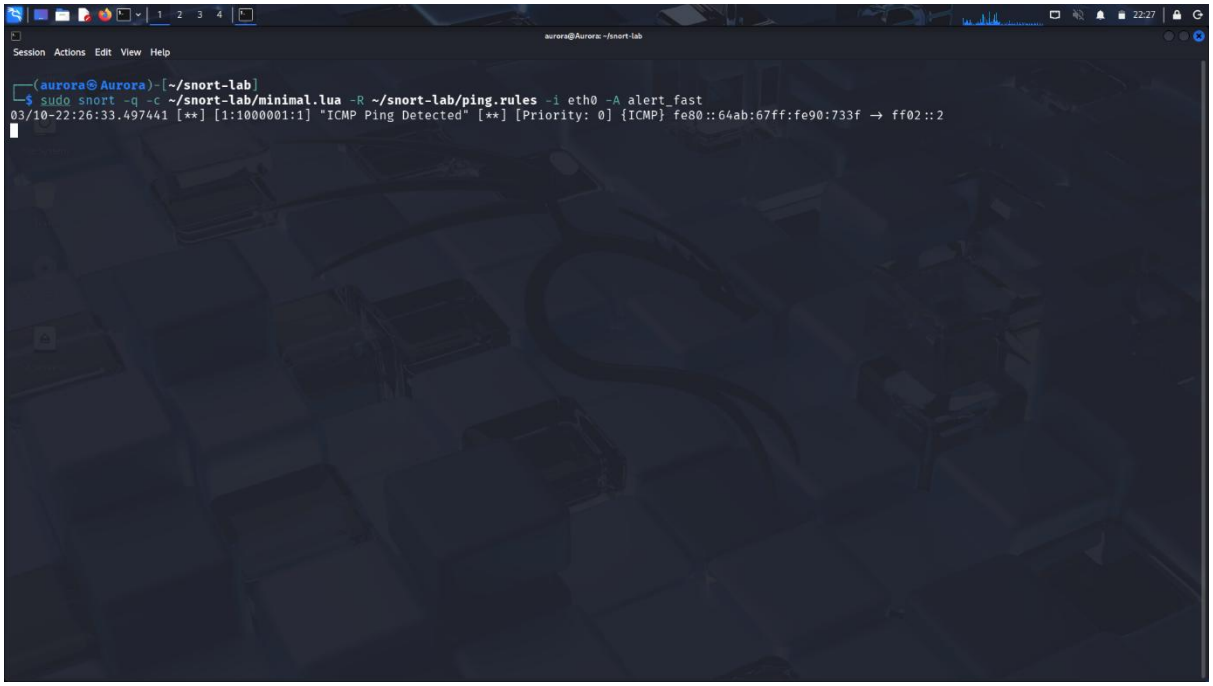

Figure 14: Ping detected by Kali VM

3.3. Use of Nessus / vulnerability scanning

Vulnerability scanning was included in the broader practical study to understand how weaknesses can be identified before exploitation and how asset exposure contributes to security risk. Tools such as Nessus can detect missing patches, insecure services, outdated software, and known vulnerabilities across hosts.

Although vulnerability scanning is different from continuous event monitoring, it supports SOC work by helping analysts understand which systems are more exposed and why certain alerts may be more critical on vulnerable assets. This practical component helped connect asset risk with monitoring priorities.

3.4. Use of Osquery / endpoint monitoring

Endpoint monitoring tools such as Osquery provide detailed visibility into host activity by allowing analysts to inspect running processes, active users, scheduled tasks, open ports, installed software, and other system-level information. In SOC work, this kind of visibility is useful during investigation because it allows suspicious behaviour to be validated directly on the endpoint.

The study of endpoint monitoring helped demonstrate how analyst visibility improves when event-based detection is complemented by direct host-state inspection. It also reinforced the principle that incident investigation often requires more than alert review alone.

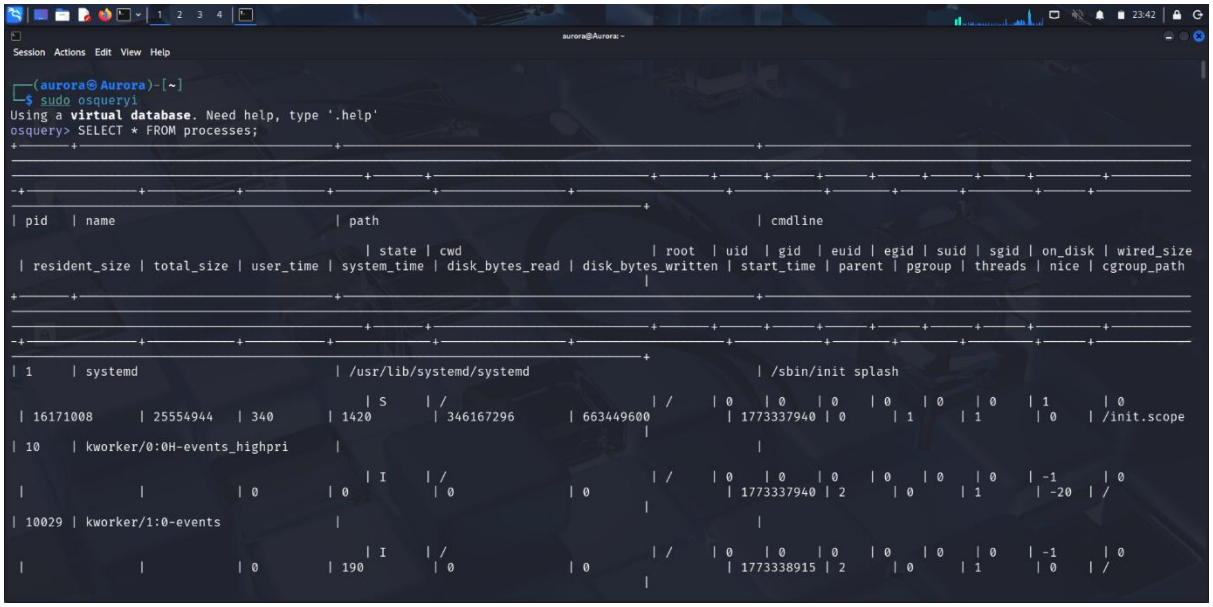

Figure 15: Endpoint monitoring output illustrating host-level visibility during investigation practice.

Part IV. Results, Observations, and Learning Outcomes

Section 7. Results and Discussion

1. Results Obtained

1.1. Findings from log analysis

The log analysis activities provided practical insight into how security-relevant events can be identified and interpreted in a SOC context. Through the review of host and system events, it became clear that logs are one of the most important evidence sources in cybersecurity monitoring. Authentication failures, service-related changes, and repeated event patterns can reveal suspicious behaviour that may not be obvious from general system usage alone. The exercise showed that even simple event analysis can help distinguish between isolated operational noise and activity that deserves closer attention.

1.2. Findings from monitoring dashboards

The monitoring dashboard activities showed that visual representations of security data significantly improve situational awareness. Instead of reviewing isolated raw logs, dashboards made it easier to identify trends, recurring event types, and active sources generating alerts or notable activity. This helped demonstrate the operational value of centralized visualization in a SOC environment, where analysts must review large amounts of information efficiently and detect important changes in near real time.

1.3. Findings from alert rule testing

The alert rule testing activities demonstrated how threshold-based detection logic can convert repeated low-level events into actionable alerts. By defining a rule that reacts when suspicious behaviour occurs multiple times within a specific time window, the monitoring system becomes more capable of detecting meaningful patterns without requiring continuous manual review. This showed the practical benefit of detection engineering in SOC work, where repeated events such as failed logins may not appear critical individually but become important when seen as a sequence.

---

2. Discussion of the Results

2.1. Value of centralized monitoring

The results obtained during the practical work highlight the importance of centralized monitoring in SOC operations. When logs, alerts, and visualizations are brought together into a single monitoring environment, analysts can review activity more efficiently and correlate events from different sources more easily. This reduces the fragmentation of analysis and improves the ability to detect suspicious behaviour in a timely manner. Centralized monitoring therefore increases visibility and supports more coherent investigation workflows.

2.2. Importance of accurate alert tuning

The results also showed that detection rules must be tuned carefully in order to be useful. If a rule is too sensitive, it may generate excessive alerts and create unnecessary noise. If it is too restrictive, it may fail to identify suspicious behaviour that should have been detected. This balance is one of the most important practical

lessons in SOC operations, because alert quality directly affects triage efficiency, analyst workload, and the ability to detect real threats.

2.3. Limits of manual analysis vs automated detection

The practical activities also highlighted the limits of purely manual analysis. Reviewing raw logs one by one can be informative for learning, but it quickly becomes inefficient when event volume increases. Manual analysis is useful for understanding how logs are structured and for validating specific cases, but it is not sufficient on its own for continuous monitoring in a real SOC environment. This is why automated detection, dashboards, and centralized search tools are essential for modern security operations.

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3. Challenges Encountered

3.1. Technical challenges

Several technical challenges were encountered during the learning and practical phases of the task. One of the main difficulties involved working across multiple components of the lab environment, including virtual machines, dashboard access, and the interpretation of event data coming from different contexts. Setting up or understanding the relationship between generated activity, collected logs, and displayed monitoring results required careful attention. In addition, some tasks required familiarity with tool interfaces and system event structures that were initially new and required gradual exploration.

3.2. Analytical challenges

In addition to technical issues, analytical challenges were also encountered. One difficulty was learning how to distinguish between isolated events and patterns that might indicate suspicious behaviour. A single event is not always meaningful by itself, and understanding whether activity is benign, abnormal, or potentially malicious requires context. This made it necessary to move beyond reading event entries individually and begin thinking in terms of frequency, correlation, and behavioural interpretation.

3.3. Learning challenges

As this task involved both theory and practical work, a learning challenge was balancing conceptual understanding with tool usage. Some concepts such as alert tuning, event interpretation, incident workflow, and SOC roles became clearer only after being connected to practical examples. At the beginning, the theoretical content could appear broad, but the lab activities helped make these ideas more concrete and operational.

---

4. Skills and Knowledge Gained

4.1. Technical skills developed

The task contributed to the development of several technical skills related to security monitoring and SOC operations. These included basic log analysis, understanding event identifiers, reviewing dashboard-based visualizations, and interpreting simple detection logic. Exposure to the lab environment also improved familiarity with how monitoring tools, virtualized systems, and event-driven workflows interact in practice. Even when certain tools were studied at an introductory level, the task helped

establish a clearer understanding of their function in a defensive security environment.

4.2. Analytical skills developed

In addition to technical knowledge, the task helped develop analytical skills required for security monitoring. These included learning how to interpret repeated events, recognize suspicious patterns, connect observations across data sources, and think about why a particular activity might deserve investigation. The exercise showed that effective SOC work depends not only on collecting data but also on making reasoned judgments based on context, frequency, and relevance

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4.3. Documentation and reporting skills developed

The task also strengthened documentation and reporting skills, which are essential in professional SOC operations. Recording observations in a structured format, summarizing findings clearly, and linking evidence to interpretation are important for communication, escalation, and later review. Through the event documentation exercise and the preparation of this report, it became clear that technical findings must be organized in a way that is understandable, traceable, and useful to others.

Part V. Conclusion and References

Section 8. Conclusion

1. General Conclusion

1.1. Summary of the work carried out

This report presented the theoretical study and practical exploration carried out during the SOC internship task on security operations, log analysis, monitoring dashboards, detection logic, and incident response fundamentals. The work combined conceptual learning with practical activities performed in a laboratory environment, allowing the understanding of how a **Security Operations Center** functions and how security events can be collected, reviewed, interpreted, and documented. Through this approach, the task provided a structured introduction to the operational logic of cybersecurity monitoring and event-driven analysis.

1.2. Importance of the task in SOC learning

This task was important in the context of SOC learning because it introduced the essential relationship between data, detection, analysis, and response. It showed that security operations are not limited to the use of tools alone, but depend on the combination of monitoring processes, analytical thinking, and structured documentation. By studying both the theory and practice of SOC work, the task helped establish a foundational understanding of how analysts identify suspicious activity, prioritize events, and support incident handling.

1.3. Overall contribution to internship experience

Overall, this task contributed significantly to the internship experience by developing a clearer understanding of how cybersecurity monitoring works in practice. It improved familiarity with SOC concepts, introduced practical workflows for reviewing and interpreting security events, and strengthened the ability to think in a more structured and analytical way about incident detection and response. The task also highlighted the importance of combining technical observation with documentation and reporting, which are essential aspects of professional security operations.

---

2. Recommendations

2.1. Suggested improvements for future practice

For future practice, it would be beneficial to extend the laboratory exercises with a more complete and repeatable workflow from event generation to alert validation and documented incident review. This would strengthen the connection between theoretical knowledge and practical monitoring activity by ensuring that each simulated event can be traced clearly from its origin to its representation in the monitoring platform. Such an improvement would also help reinforce the understanding of log pipelines, event visibility, and the full alert lifecycle.

2.2. Further areas to explore in SOC operations

As a continuation of this learning path, several areas of SOC operations could be explored in greater depth. One important area is advanced detection engineering, including the design of more refined correlation rules, behavioural alerts, and better threshold selection. Another useful area is incident investigation methodology, particularly the correlation of host, network, and user activity in order to build stronger event timelines and improve decision-making during triage.

Section 9. References and Appendices

1. References

Standards and frameworks

The theoretical and methodological foundation of this report is supported by recognized cybersecurity standards and frameworks commonly used in security operations and incident response. These references provide structured guidance for understanding monitoring, detection, incident handling, and attacker behaviour. In particular, standards related to incident response and frameworks used for threat classification are essential for building a consistent SOC-oriented approach.

1. **National Institute of Standards and Technology** (NIST), *Computer Security Incident Handling Guide*, Special Publication 800-61.
2. **National Institute of Standards and Technology** (NIST), *Cybersecurity Framework (CSF)*.
3. **MITRE ATT&CK**, *Adversarial Tactics, Techniques, and Common Knowledge Framework*.
4. **Elastic**, *Elastic Security and Kibana Documentation*.
5. **Wazuh**, *Official Documentation*.
6. **Microsoft**, *Windows Event Viewer and Windows Security Event Documentation*.
7. **Snort**, *Official Documentation and Rule Writing Guide*.
8. **Tenable**, *Nessus Essentials Documentation*.
9. **Osquery**, *Official Documentation*.

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